

Unified Deep Learning Model for Multitask Reaction Predictions with Explanation

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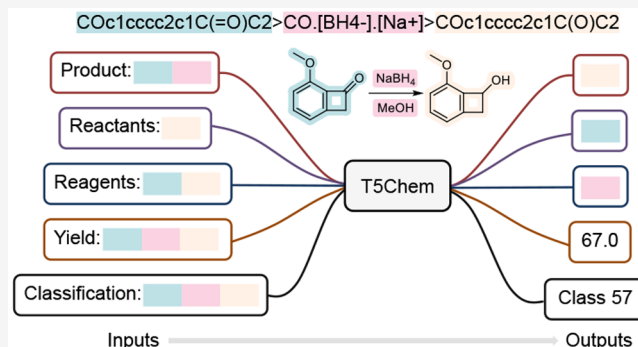


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ABSTRACT: There is significant interest and importance to develop robust machine learning models to assist organic chemistry synthesis. Typically, task-specific machine learning models for distinct reaction prediction tasks have been developed. In this work, we develop a unified deep learning model, T5Chem, for a variety of chemical reaction predictions tasks by adapting the “Text-to-Text Transfer Transformer” (T5) framework in natural language processing (NLP). On the basis of self-supervised pretraining with PubChem molecules, the T5Chem model can achieve state-of-the-art performances for four distinct types of task-specific reaction prediction tasks using four different open-source data sets, including reaction type classification on USPTO_TPL, forward reaction prediction on USPTO_MIT, single-step retrosynthesis on USPTO_50k, and reaction yield prediction on high-throughput C–N coupling reactions. Meanwhile, we introduced a new unified multitask reaction prediction data set USPTO_500_MT, which can be used to train and test five different types of reaction tasks, including the above four as well as a new reagent suggestion task. Our results showed that models trained with multiple tasks are more robust and can benefit from mutual learning on related tasks. Furthermore, we demonstrated the use of SHAP (SHapley Additive exPlanations) to explain T5Chem predictions at the functional group level, which provides a way to demystify sequence-based deep learning models in chemistry. T5Chem is accessible through <https://yzhang.hpc.nyu.edu/T5Chem>.



INTRODUCTION

Organic synthesis is one of the most fundamental problems in chemistry. With the advances of computing power, data availability, and algorithms, there has been significant interest in developing machine learning (ML) models to assist a variety of organic reaction-related tasks,^{1–4} including reaction product prediction,^{5–18} retrosynthesis,^{9,14,17,19–37} reaction type optimization,^{38–41} reaction yield prediction,^{42–54} and reaction type classification.^{38,51,55–57} These ML-based data-driven approaches for organic synthesis can be classified into descriptor-based models,^{5–10,19–26,38–41,51,55,57} graph-based models,^{11–13,27,28,52} and sequence-based models,^{14–18,29–37,53,54,56} depending on how molecules are represented as input for machine learning. Descriptor-based models use hand-crafted features as molecular representations and often need feature engineering or template extraction for different reaction prediction tasks, which set limitations to generalizability.^{43,58,59} For this reason, people start to turn to end-to-end models. Graph-based models treat molecules as graphs where atoms are viewed as nodes, and bonds are viewed as edges. Although molecules can be naturally represented as graphs, stereoisomers, molecular structures with the same graph connectivity but different spatial arrangements, remain underexplored despite few previous works.^{60,61} Furthermore, most graph-based models suffer from the need of atomic

mapping and inability of handling stereoisomers. Currently few attempts have been made to develop graph-based models for reaction yield prediction and reaction type prediction. On the other hand, sequence-based models showcased the feasibility of language modeling strategy on chemical reactions by using text-based representations of molecular graphs. Simplified molecular-input line-entry system (SMILES⁶²) is a commonly used^{63,64} linear string notation in deep neural networks. Accordingly, reaction prediction can be formulated as a machine translation problem, where reactants/reagents SMILES serve as the source language, and product SMILES serve as the target sequences.

The recent key advance that has been made for deep learning in language modeling is the transformer,⁶⁵ which stands out for its outstanding performance and wide applicability. Schwaller et al.¹⁶ adapted the transformer architecture for the forward reaction prediction task, namely,

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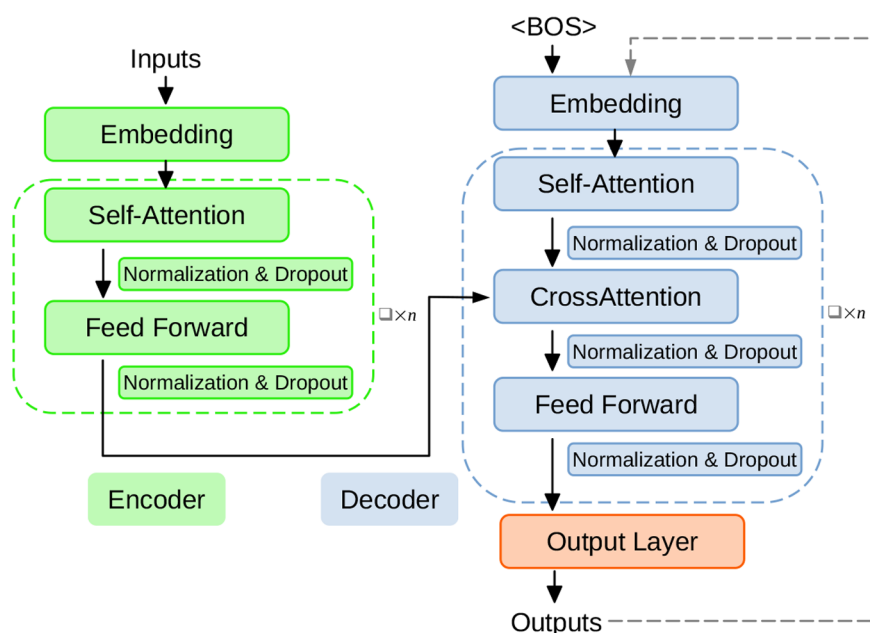


Figure 1. Illustration of the general transformer architecture⁶⁵ used in T5⁶⁹ and T5 Chem.

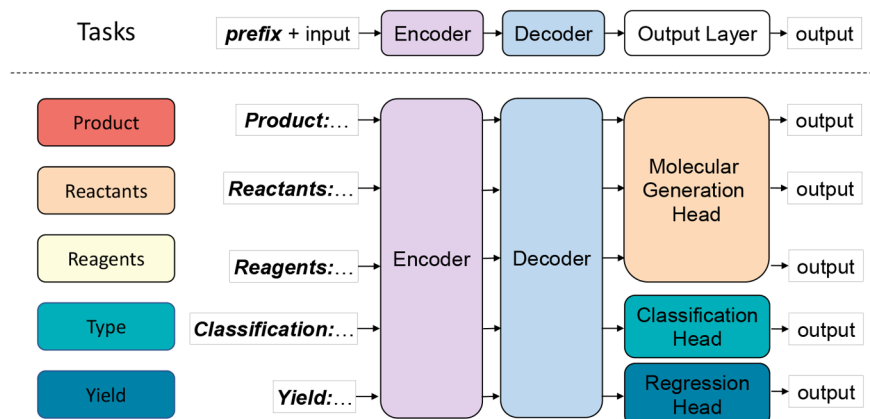


Figure 2. Multitasking of T5Chem. Five different tasks are shown. All models have a general structure of encoder, decoder, and output layers. However, they may have different prompts and head types, depending on task types.

molecular transformer, for the first time. After that, many efforts have been made to apply transformer models for the retrosynthesis task.^{14,17,30–36} Recently, Schwaller et al. developed a new type of molecular fingerprint, rxnfp,⁵⁶ for reaction type classification. The rxnfp is derived from bidirectional encoder representations from transformers (BERT),⁶⁶ a transformer-based encoder that converts an input sequence into an internal vector representation. The BERT classifier reached a classification accuracy of 98.2% in an open-sourced data set. Soon afterward, they extended the BERT model with a regression layer to predict reaction yield.⁵³

Transformer-based models not only shed light on general-purpose reaction predictions but are also transferable to particular reactions. Pesciullesi et al.¹⁸ published a transfer learning approach to predict regioselective and stereoselective reactions on carbohydrates with molecular transformer. Moreover, transformers take advantage of pretraining^{56,67,68} and data augmentation¹⁷ with improved performance. In spite of all the advances transformers have made, models developed

for different reaction prediction tasks are still not interchangeable.

Herein, we present a unified deep learning model, T5Chem, for a variety of chemical reaction predictions tasks by adapting the Text-To-Text Transfer Transformer⁶⁹ (T5). The T5 model is structurally similar to the original Transformer,⁶⁵ except that instead of using the Layer Norm bias, it places a layer normalization outside the residual path and uses a different, relative position embedding scheme. The idea of multitasking predictions is inspired by the natural human learning process that knowledge learned from one task should be helpful to other related tasks. Meanwhile, we introduce a new multitask reaction prediction data set, USPTO_500_MT, which can be used to train and test five different types of reaction tasks, including a novel reagent suggestion task.

In the following, we show how T5Chem can be tailored to tackle multiple reaction prediction tasks with task-specific prompts, which requires no or minor modifications on output layers. Then, we apply this framework to several open-source data sets individually to illustrate its applicability and state-of-

the-art performances. After that, we demonstrate that this unified T5Chem model can be trained on the new unified multitask reaction prediction data set USPTO_500_MT without degrading performance on individual tasks. Finally, we look into specific prediction examples and shed light on black-box model predictions with SHAP⁷⁰ (SHapley Additive exPlanations).

METHOD

Model. The T5Chem model is developed based on Text-To-Text Transfer Transformer⁶⁹ (T5), which is an encoder–decoder model from the transformer family, as illustrated in Figure 1. In comparison with the original transformer model,⁶⁵ the T5 model uses the same self-attention strategy with two modifications: (1) It removes the layer norm bias and places the normalization outside of the residual path. (2) It uses a relative positional embedding scheme instead of sinusoidal position embedding. The T5 model is designed to convert all language problems into a text-to-text format so that it is capable of working on a variety of tasks at the same time. To specify which task the model should perform, a task-specific prompt would be prepended to the original input sequence before feeding it to the model.

To develop the unified deep learning model T5Chem for a variety of chemical reaction predictions tasks, besides the use of character-level tokenization for the SMILES input and introduction of task-specific prompts for different reaction prediction tasks into the vocabulary as special tokens, we have modified the original T5 output layer (language modeling head) into three different kinds in T5Chem as illustrated in Figure 2: the molecular generation head for all sequence to sequence tasks, i.e., reaction product prediction, single-step retrosynthesis, and reagent suggestion; the classification head for reaction type classification; and the regression head for product yield prediction which employs the soft-label strategy.

Tokenization is the process of breaking a sequence into small chunks. In Natural Language Processing (NLP), tokenization can be performed on either word or subword level. Similarly, different tokenization approaches can be applied to molecules. T5Chem employs character-level tokenization, which simply splits reaction SMILES into a single alphabet letter, digit, or special symbol. In comparison with atom tokenization which runs a WordPiece tokenization algorithm over SMILES strings using a regular expression¹⁵ and SELFIES which is short for Self-Referencing Embedded Strings,⁷¹ one main advantage of the character-level tokenization is its simplification, flexibility, and much smaller vocabulary size. In our exploration, neither atom tokenization nor SELFIES is found to outperform character-level tokenization for reaction prediction tasks (Table S1).

In T5Chem, each task-specific prompt is an English word that would not appear in original SMILES and thus would be added to the vocabulary as a special token. Here we use “Product:” for reaction product prediction, “Reactants:” for single-step retrosynthesis, “Reagents:” for reagent suggestion, “Classification:” for reaction type prediction, and “Yield:” for reaction yield prediction. Note that special tokens only work as identifiers for different tasks, and the exact wording of these prompts would not impact our model training.

For chemical reaction predictions, different tasks can have different label formats, and thus, the output layer needs to be changed accordingly, as illustrated in Figure 2. For sequence-to-sequence tasks like forward reaction prediction, retrosyn-

thesis, and reagents prediction, the input and output sequences share the same vocabulary. Correspondingly, the output layer, molecular generation head, shares weights with the input embedding layer and produces a probability distribution among the whole vocabulary space.

For classification tasks such as reaction type classification, a classification head was used as the output layer. This is a new linear layer that outputs a probability distribution among all classification categories. As an example, for a reaction type classification task consisting of 1000 reaction template classes, the classification head for this task would be a linear layer with an output size to be 1000.

For regression tasks such as reaction yield prediction, we applied the soft label strategy for min–max normalized training labels; the regression head is a single linear layer with an output dimension to be 2, representing weights for the min and max values. Different from molecular generation and classification heads, the Kullback–Leibler divergence is used as the loss function to minimize the probability distribution between outputs and true labels for regression tasks.

Implementation and Training Details. T5Chem is implemented in Python (version 3.7) by using RDKit (version 2021.03.1)⁷² for reaction preprocessing and SMILES validation and employing Pytorch (version 1.7.0)⁷³ and huggingface transformers (version 4.10.2)⁷⁴ for seq2seq modeling. The source code is available online at <https://github.com/HelloJocelynLu/t5Chem>.

Similarly to the original work, a BERT⁶⁶-like “masked language modeling” objective would be used for model pretraining in a self-supervised manner. Here, 97 million molecules from Pubchem⁷⁵ were used for pretraining. During the pretraining, tokens of source sequences would be randomly masked, and the goal is to predict correct tokens that have been masked. We applied the same mask rate, 15%, as the original T5 model. Masked tokens would be substituted to the mask token (<mask>) 80% of the time or be replaced by another random token in the vocabulary 10% of the time and remain the same for the rest of the time. Then, the model would be fine-tuned in supervised downstream tasks. During the fine-tuning, various task-specific prompts and output layers would be used for different output styles, as illustrated in Figure 2.

In T5Chem, the molecular generation head for all sequence to sequence tasks, including reaction product prediction, reactants prediction, and reagents prediction, is the same as the output layer of T5ForConditionalGeneration with the huggingface transformers⁷⁴ package. It produces a probability distribution among the same vocabulary space as the input sequence. For classification tasks, the classification head is used, which is a linear output layer that generates a probability distribution among all classification categories. Both the molecular generation head and classification head use cross entropy loss as the loss function. On the other hand, for regression tasks, the soft label strategy is used for min–max normalized training labels, and the regression head is a single linear layer with an output dimension of 2, representing the probability distribution between min and max. The Kullback–Leibler divergence is used as the loss function to minimize the probability distribution between outputs and true labels for regression tasks. During the test phase, the molecular generation head of T5Chem repeatedly generates molecules token by token, until the “end of sentence token” is generated or the maximum length of the allowed prediction is reached,

Table 1. Data Set Splits Used for the Experiments

Data set	Train	Valid	Test	Total	Task
USPTO_TPL ^{56a}	360,545	40,059	44,511	445,115	Reaction type classification
USPTO_MIT ¹²	409,035	30,000	40,000	479,035	Forward prediction
USPTO_50k ^{29a}	40,029	5004	5004	50,037	Retrosynthesis
C–N Coupling ^{44a,b} (Random splits)	2767	–	1188	3955	Reaction yield prediction
C–N Coupling ^{44a,b} (Out-of-sample test 1)	3057	–	898	3955	Reaction yield prediction
C–N Coupling ^{44a,b} (Out-of-sample tests 2, 4)	3055	–	900	3955	Reaction yield prediction
C–N Coupling ^{44a,b} (Out-of-sample test 3)	3058	–	897	3955	Reaction yield prediction
USPTO_500_MT ^a	116,360	12,937	14,238	143,535	Multitask prediction

^aContains stereochemical information. ^bWith reactants/reagents separation.

while both classification and regression heads can be viewed as a one-time generation procedure which has the maximum allowed prediction length to be 1.

For all tasks, T5Chem uses a whole encoder and decoder architecture with four layers and eight attention heads. We used 256 as the hidden dimension for T5Chem and 2048 for the intermediate feed forward layer. The maximum vocabulary size was set to 100, which includes 70 tokens from character-level tokenization of SMILES of pubchem and USPTO molecules, five special structural tokens (<s>, </s>, <unk>, <pad>, <mask>), six prompting tokens, and 19 placeholders. The <pad> token is used as the first input token for the decoder to initialize the decoding process. It is worth mentioning that placeholders can be replaced for other prompting tokens at any time when needed, which makes T5Chem extendable to more tasks without retraining the whole neural network.

DATA SET

In this work, we used four open-sourced data sets introduced by previous published work for better comparison. We also introduced one new reaction data set, USPTO_500_MT, for multitasking purposes. All chemical reactions were described using the SMILES representation as inputs for T5Chem. Table 1 shows an overview of these data sets.

USPTO_TPL for Reaction Type Classification. The data set for the reaction type classification task was originally derived from the USPTO database by Lowe⁷⁶ and introduced by Schwaller et al.⁵⁶ This strongly imbalanced data set consists of 445,000 reactions divided into 1000 classes without reactants–reagents separation. Reaction classes here were defined by SMART reaction templates and obtained by atom-mapping the USPTO data set with RXNMapper. Among all USPTO reactions, the 1000 most frequent template hashes were selected as targets. Reactions were randomly split into 90% for training and validation and 10% for testing.

USPTO_MIT for Forward Reaction Prediction. This benchmark data set has been prepared by Jin et al.¹² based on the USPTO database of Lowe⁷⁶ and includes both separated and mixed versions. The mixed version without reactants–reagents separation is used as a more challenging task which needs to learn to identify “reagents” and “reactants” by themselves. In machine-aided reaction prediction, reagents are defined as chemical species that do not appear in (major) products. Therefore, the mixed version reflects a real-world scenario where users do not have prior knowledge about

products before predictions were made. This data set consists of 479,000 reactions: 409,000 for training, 30,000 for validation, and 40,000 for testing.

USPTO_50K for Single-Step Retrosynthesis. Retrosynthesis is the process of deconstructing a molecule. More specifically, the single-step retrosynthesis prediction task is defined as given a product as input, it finds reactant combinations that could generate the input compound. The data set was a filtered version of Lowe’s patent data set. It contains only 50,000 reactions that have been classified into 10 broad reaction types.⁷⁷ In this work, we did not remove stereochemical information and did not provide a model with reaction types as we would encounter in real-world situations. We followed the splitting proposed by Liu et al.²⁹ that also has been used in many works.^{17,19,27,28,30,31} Here, 40,000, 5000, and 5000 reactions were used for training, validation, and testing, respectively.

C–N Coupling Data Set for Reaction Yield Prediction.

This is a high-throughput experimental data set. In 2018, Ahneman et al. performed 4608 Pd-catalyzed Buchwald–Hartwig C–N cross coupling reactions.⁴³ These nanomole-scale experiments were carried out on three 1536-well plates consisting of a full matrix of 15 aryl and heteroaryl halides, three bases, four ligands, and 23 isoxazole additives. Here, 3955 applicable reactions after removing control groups were used. In their work, a random forest model was used with 120 DFT calculated features of compounds. This strategy gave a promising performance. Recently, Schwaller et al.⁵³ proposed a BERT-based reaction encoder enriched with a regression layer (Yield-BERT). This model can directly take reaction SMILES as input and thus is extendable to other reaction types without feature engineering. It outperforms the random forest model on many tests except for the most challenging out-of-sample test. To demonstrate the capability of T5Chem on regression tasks, we followed previous data splitting: 10 random splitting (70/30 for training/testing) and four out-of-sample sets. Test samples in out-of-sample sets contain reactions additives which are not included in the training data, which can be used to evaluate the generalizability of machine learning models.

USPTO_500_MT for Multitask Reaction Prediction. In order to illustrate the multitasking capability of our unified model, we introduced a new data set that is applicable for multiple reaction prediction tasks, including forward reaction prediction, reactants prediction (single step retrosynthesis), reagents prediction, reaction yield prediction, and reaction type classification. The same splitting is expected for all tasks to

avoid a possible data leakage problem. A more detailed overview for this newly introduced data set is available in Figures S1 and S2 of the Supporting Information.

USPTO_TPL (in Task 1: Reaction Type Classification). This is a good starting point to construct this multitasking data set, as it contains reactants, reagents, products, and reaction types for every reaction instance. In order to recover reaction yield for each reaction, we looked into text-mined records⁷⁸ by Lowe and matched reactions based on reactants, reagents, and products. In the original record, yields are reported either directly mined from the document or calculated from the isolated product amount or both. However, discrepancies exist between text-mined and calculated values for a same reaction in the same patent,³ and some calculated yields have values greater than 100%. Thus, all yields with values greater than 100% were removed first. Then, for reactions with only one type of yield available, that one is used as the label for reaction yield. When both text-mined and calculated values are available, their average value is used if the difference between the two values is less than 10%; otherwise, the text-mining value is used. A successful curation of reaction yield with a complete match is only achieved for about one-third of the reactions in the USPTO_TPL data set. We noticed that some reaction classes are very sparse (as many reactions cannot be recovered). Therefore, we kept the top 500 most frequent reaction classes and ended up with 116,000 reactions for training, 13,000 reactions for validation, and 14,000 reactions for testing. We refer this data set as USPTO_500_MT.

RESULTS AND DISCUSSION

One of the most intriguing features for transformer models is the effectiveness of its pretraining. T5Chem was pretrained with a masked language modeling (MLM) objective on 97 million PubChem molecules. The pretrained model shows a steady learning curve and faster convergence (Figure S3), and it has been used to initialize all models used in the following experiments. First, we examined T5Chem for four distinct types of task-specific reaction prediction tasks using four different open-source data sets, and the results indicated that the T5Chem model can achieve state-of-the-art performances. Then, we applied it to USPTO_500_MT, which demonstrates the multitasking capability and robustness of T5Chem. Finally, we used SHAP (SHapley Additive exPlanations) to explain T5Chem predictions at the functional group level, which illustrates how to demystify sequence-based deep learning models in chemistry.

Task 1: Reaction Type Classification. Here, we used the USPTO TPL data set the same way as the original work, i.e., the BERT classifier,⁵⁶ in which reactants and reagents are not separated. T5Chem has been trained for 100 epochs, and the test results are summarized in Table 2. Our model outperformed the BERT classifier⁵⁶ and achieved 99.5% accuracy.

Table 2. Results for Reaction Type Classification^a

Model	Accuracy	CEN	MCC
BERT classifier ⁵⁶	0.989	0.006	0.989
T5Chem	0.995	0.003	0.995

^aThe lower the confusion entropy of a confusion matrix⁷⁹ (CEN) and the higher the Matthews correlation coefficient⁸⁰ (MCC) the better. Bold entries highlight the best-performing approach.

Looking at the confusion entropy of a confusion matrix (CEN) and the overall Matthews correlation coefficient (MCC) also leads to the same conclusion.

Task 2: Forward Reaction Prediction. USPTO_MIT has been used for benchmarking forward predictions in many previous works. T5Chem was trained on this data set with a task-specific prompt “Product” for 30 epochs. The results are summarized in Table 3 with comparison to some previous models. Seq2seq¹⁵ is one of the first attention-based sequence-to-sequence model that treats molecules as strings. WLDN5¹³ performed as a two-stage model using a graph-convolutional neural network. Both models were only evaluated with reactants–reagents separation. We presented results as top-k accuracy. Top-k accuracy takes k model predictions with the highest probability. If one of them is a true label, it classifies the prediction as correct. Note that here we just used a single model without any data augmentation. T5Chem shows slightly better performance in comparison with a molecular transformer¹⁶ and clearly outperforms both Seq2seq and WLDN5.

Task 3: Single-Step Retrosynthesis. In this task, we used the same model architecture as the forward reaction prediction. We trained T5Chem on USPTO_50k for 100 epochs without given reaction types and compared our results with some previous studies with sequence-based models using the same training data. The results for T5Chem without data augmentation are shown in Table 4. Specifically, the Seq2seq²⁹ model utilizes LSTM⁸¹ architecture. Molecular Transformer¹⁴ was based on the Transformer architecture⁶⁵ which was solely based on self-attention mechanisms. SCROP³² is short for self-corrected retrosynthesis predictor; they used a transformer-based retrosynthetic reaction predictor coupling with a neural network-based syntax corrector. We can see that T5Chem is capable of performing well for retrosynthesis as a single task, as it achieves better performance in comparison with other smiles-based sequence-to-sequence models.

Task 4: Reaction Yield Prediction. In order to fit our T5Chem to reaction yield prediction, a regression head was added to give number outputs as described in the Method section. Task-specific prompt “Yield” was prepended to input reaction SMILES to specify the task type. To test the proposed approach, the same splits were used as in Sandfort et al.⁴⁴ and Schwaller et al.⁵³ We trained T5Chem on this cross-coupling data for 100 epochs. The results are shown in Table 5.

Since only reaction SMILES were used in our model, this approach can be easily adapted to other reaction types without any modifications or feature engineering. Generally, our model performs best among listed models. For random splitting, our model achieved an average R^2 of 0.970. For more challenging out-of-sample tests, T5Chem also gets best results except for Test 1. Test 4 is viewed as the most challenging test among all test sets as shown in other models. This might imply that T5Chem would have better generalizability in this task. To further investigate our model predictions on four test sets, we examined the prediction accuracy for each test. In a real-world problem like singleton batch experiments and THE, there are underlying experimental errors in measurements. To address this issue, we used $\pm 10\%$ as acceptable error range and calculated prediction accuracy within this range as shown in Table 6.

Task 5: Multitask Prediction. After showing that T5Chem is able to achieve comparable or better performance with other sequence-based models for individual tasks, we carried out multitask experiments with the newly prepared

Table 3. Results for Forward Reaction Prediction^a

Model	Mixed			Separated		
	Top 1 (%)	Top 2 (%)	Top 5 (%)	Top 1 (%)	Top 2 (%)	Top 5 (%)
Seq2seq ¹⁵	—	—	—	80.3	84.7	87.5
WLDNS ¹³	—	—	—	80.6	90.5	93.4
Molecular Transformer ¹⁶	88.6 ^b	92.4 ^b	94.2 ^b	88.8	92.6	94.4
T5Chem	88.9	92.9	95.2	90.4	94.2	96.4

^aT5Chem achieved comparable performance with the molecular transformer model on this individual task. Bold entries highlight the best-performing approach. ^bIt is unclear whether the data augmentation strategy has been used for these results.

Table 4. Results for Single-Step Retrosynthesis on USPTO_50k with Other Smiles-Based Sequence-to-Sequence Models^a

Model	Top 1 (%)	Top 3 (%)	Top 5 (%)
Seq2seq ²⁹	37.4	52.4	57.0
Molecular Transformer ¹⁴	43.5	60.5	—
SCROP ³²	43.7	60.0	65.2
T5Chem	46.5	64.4	70.5

^aBold entries highlight the best-performing approach.

USPTO_500_MT data set to demonstrate multitasking capability and transferability of T5Chem. The new data set USPTO_500_MT contains five objectives: forward reaction prediction, single-step retrosynthesis, reagent prediction, reaction type prediction, and reaction yield prediction. Note that the curation of this data set does rely on atom-mapping for labeling, but T5Chem does not require atom-mapping as inputs. The training, validation, and testing sets are well separated to ensure no reaction overlapping across all task types.

We first applied T5Chem in all tasks independently to get baseline results. Although the model has already achieved promising results, we are interested in figuring out whether further performance gains can be obtained by cross-task training. The five tasks are grouped into two subgroups. The first subgroup includes forward reaction prediction, single-step retrosynthesis, and reagents prediction, as they are all sequence-to-sequence tasks and can share the same model architecture. The second subgroup consists of reaction classification and reaction yield prediction. Both tasks take the whole reaction sequence as inputs, and we proposed that the model can be trained on the two tasks at the same time by combining their loss functions together. A multitask training using combined loss functions from all five tasks has also been carried out with worse performance. It may due to the fact that the regression and classification tasks are internally less similar than sequence to sequence predictions. When we trained them together, we forced the major part (embedding, encoder, and

Table 6. Accuracies for Reaction Yield Prediction on C–N Coupling Test Sets^a

	Test 1	Test 2	Test 3	Test 4
Accuracy (%)	75.5	83.9	64.5	59.9

^aPredictions within 10% error range are viewed as accurate.

decoder) to share the same weights among all tasks. The molecular generation head also shares weights with embedding layer. The only flexibility components, regression head and classification head, are not sufficient to differentiate those tasks. The results can be found in Table S2 of the Supporting Information.

For the sequence-to-sequence subgroup, we mixed training sets of three tasks. To distinguish different tasks, task-specific prompts of “Product”, “Reactants”, and “Reagents” were used. We used a combined, big training set as shown in Figure 3. In addition, we explored 5-fold data augmentation using non-canonical SMILES and found that data augmentation did not improve performance on the USPTO_500_MT data set in terms of top-k accuracy (even slightly worse results), but it did improve SMILES validity (Table S3).

In this experiment, we only trained one model with the mixed data set and then tested it separately on three test sets from different tasks. We compared this result with individual models that were trained and tested on separate data sets. Two metrics have been reported: the accuracy in top-k predictions and the invalid SMILES rate in top-k predictions. Both metrics are calculated on all predictions. The valid SMILES is defined as a SMILES string that can be parsed by the RDKit package. Note that one may have a high top-k accuracy and high SMILES invalidity at the same time. For example, if T5Chem gives five predictions to a custom input, and the first prediction is correct. Then, all top-k accuracy would be 100% because the first prediction is always included. However, if the second–fifth predictions are wrong and chemical invalid, we have top-5 SMILES invalidity as high as 80% (4/5)! The results are summarized in Figure 4.

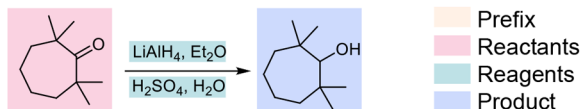
Table 5. Results for Reaction Yield Prediction on C–N Coupling Reactions^a

R ²	DFT ⁴³	MFF ⁴⁴	Yield-BERT ⁵³	T5Chem
Random 70/30	0.92	0.927 ± 0.007	0.951 ± 0.005	0.970 ± 0.003
Test 1	0.80	0.851	0.838	0.811
Test 2	0.77	0.713	0.836	0.907
Test 3	0.64	0.635	0.738	0.789
Test 4	0.54	0.184	0.538	0.627
Avg. Tests 1–4	0.69 ± 0.104	0.596 ± 0.251	0.738 ± 0.122	0.785 ± 0.094

^aBold entries highlight the best-performing approach.

- Forward reaction prediction: "Product:" + reactants/reagents + ">>" → product
- Retrosynthesis: "Reactants:" + product → reactants
- Reagents prediction: "Reagents:" + reactants + ">>" + product → reagents

Example:



prefix+CC1(C)CCCC(C)(C)C1=O>CCOCC.O.O=S(=O)(O)O.[Al+3].[H-].[H-].[H-].[Li+]>CC1(C)CCCC(C)(C)C1O

Source:

Product: C1C(C)(C)C(=O)C(C)(CCC1)C.CCOCC.O.[H-].[H-].[H-].[Al+3].O=S(=O)(O)O.[H-].[Li+]>>

...

Reactants: CC1(C)CCCC(C)(C)C1O

...

Reagents: CC1(C)CCCC(C)(C)C1=O>>CC1(C(C(C)(C)CCCC1)O)C

...

Target:

CC1(C(C(C)(C)CCCC1)O)C

CC1(C(=O)C(CCCC1)(C)C)C

CCOCC.O.O=S(=O)(O)O.[Al+3].[H-].[H-].[H-].[Li+]

Figure 3. Mix training data set from different tasks. In the multitask training scheme, we combined the forward reaction prediction task, retrosynthesis task, and reagents prediction task together. Every input instance starts with a task-specific prompt, followed by actual input. In forward reaction prediction, the model takes reactants and reagents (without separation) as a source sequence. In retrosynthesis, the model takes only product SMILES as a source sequence. In reagents prediction, the model takes both reactants and product SMILES as inputs. A reduction reaction is shown as an example.

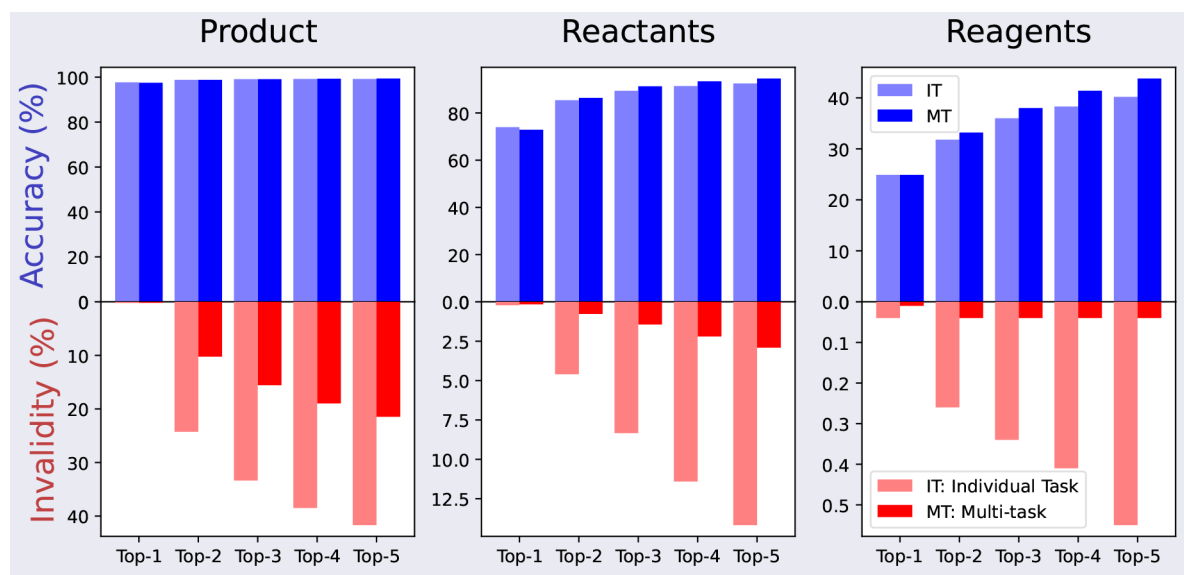


Figure 4. Test results on USPTO_500_MT. We evaluated both accuracy and SMILES invalidity for top-k predictions on different tasks. Individual tasks and multitask training schemes have comparable performance in accuracy for all three tasks. But combined training gives much lower invalid SMILES rate. Note that one may have a high top-k accuracy and high SMILES invalidity at the same time.

T5Chem achieves top-1 accuracy of 97.5%, 72.9% and 24.9% for forward prediction, retrosynthesis and reagents prediction, respectively. It achieves comparable performance as being trained on separate tasks in terms of top-k accuracy. This demonstrates that these three tasks are closely related, and are possible to learn at once. T5Chem also generates much less grammatically invalid molecules when being trained on mixed training data, especially when more molecules are generated (when $k > 1$ in top-k predictions). It may indicate that leveraging knowledge from different yet related tasks is helpful to build a more robust model.

Upon deeper look into the new task—reagents prediction task, T5Chem actually made some reasonable suggestions for

those some seemingly "wrong predictions". Figure 5 shows one reaction from test set that T5Chem failed to predict. The proposed reagents do not have the exact match and therefore being labeled as "incorrect". But similar transformation (reaction class template 112) under proposed conditions can be found in training data set.

Previously, people needed to train models separately for different tasks. By using mixed data set, we can obtain one common model that is able to do three tasks at the same time without further fine-tuning.

The other two tasks, which include reaction type classification and reaction yield prediction, take whole reaction sequence as inputs, we hypothesized that the model may

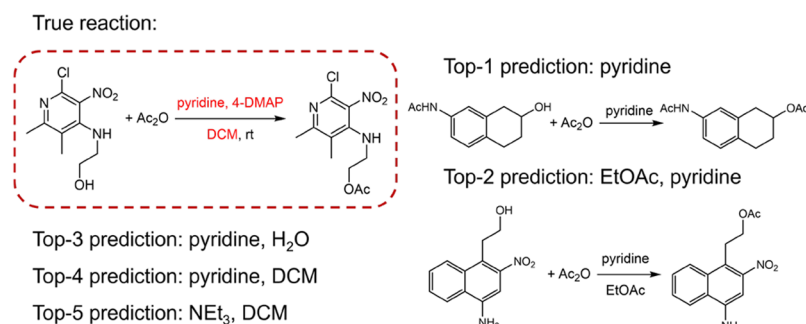


Figure 5. A alcohol protection reaction with an acetate group. The ground true answer is colored in red. T5Chem failed to predict the exact match, but the proposed predictions are also reasonable despite they are identified as "wrong predictions". Similar transformation (reaction class template 112) under the top-2 proposed conditions can be found in training data set.

Table 7. Multi-task testing results with T5Chem on USPTO_500_MT. The five tasks are grouped into two subgroups. The first subgroup includes forward reaction prediction, single-step retrosynthesis and reagents prediction, as they are all sequence-to-sequence tasks and can share the same model architecture. The second subgroup consists of reaction classification and reaction yield prediction. Both tasks take the whole reaction sequence as inputs, and can be trained at the same time by combining their loss functions together. Only one model was trained per group with the mixed dataset

Task Type	Forward	Retrosynthesis	Reagents	Classification	Yield
Metrics	Top-1 Accuracy	Top-1 Accuracy	Top-1 Accuracy	Accuracy	R
Results	97.5%	72.9%	24.9%	99.4%	0.46

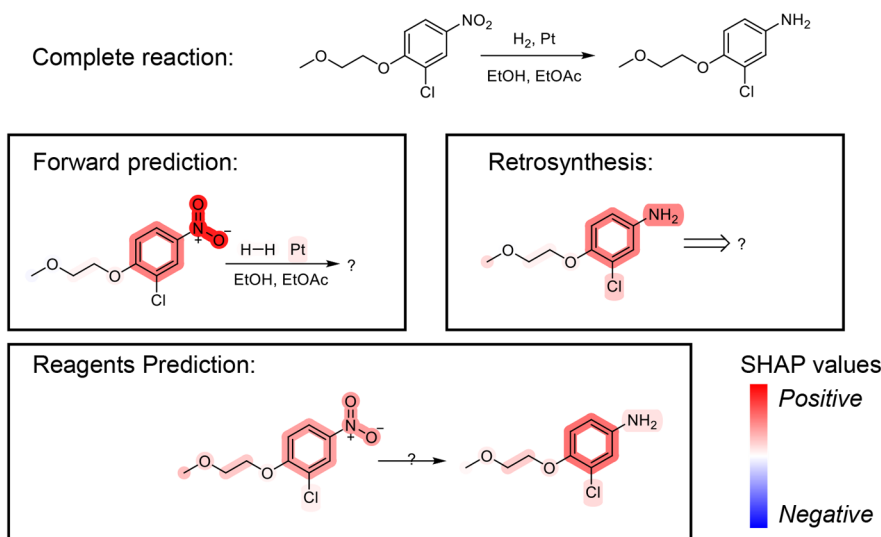


Figure 6. Visualize SHAP values in multitask predictions. The intensity of color stands for the magnitude of SHAP values. Positive contributions are drew in red and negative contributions are drew in blue.

benefit from transfer learning. We first trained a reaction type classification model as an individual task, and got an accuracy of 99.6% in reaction type classification with USPTO_500_MT. This result is close to that for USPTO_TPL as shown in Table 2. The classification head of T5Chem was then replaced by a regression head for reaction yield prediction. This transfer learning model achieves better performance on yield prediction than directly training in terms of both R^2 (0.22 v.s. 0.20) and MAE (17.5 vs 17.8) in percentage yield. Furthermore, we build a combined model to train both tasks together. During training, we calculated summation of losses from individual tasks and selected the best checkpoint based on validation loss. The combined model got an accuracy of 99.4% in reaction type classification, R^2 of 0.22 and MAE of 17.8 in reaction yield prediction. The results are comparable to its counterpart that

be trained individually. The prediction accuracy within $\pm 10\%$ absolute error range is 39.7% for this combined model.

Table 7 summarized the results of T5Chem on USPTO_500_MT. We observed promising results on this multitask data set. Note that reaction yield prediction in USPTO has always been a challenging task as there are many noises in this data set. The best model until now applies yield prediction at different mass scales, and only get R^2 of 0.117 (gram) and 0.195 (subgram).⁵³

Model interpretation with SHAP. In order to explain predictions given by our T5Chem, we applied SHAP⁷⁰ (<https://github.com/slundberg/shap>) to explain T5Chem predictions at the functional group level. SHAP is short for SHapley Additive exPlanations, which has been one of the

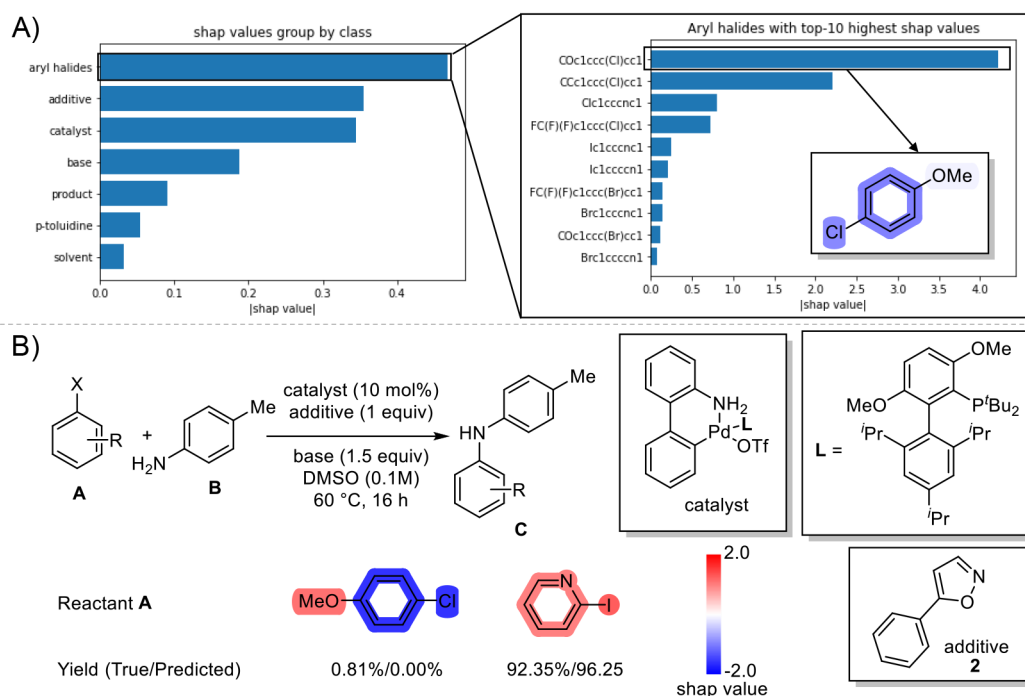


Figure 7. Visualize SHAP values in reaction yield predictions. (A) SHAP values show that the selections of aryl halides, additives, and catalysts contribute most to yield predictions. Among all compounds, 4-chloromethoxybenzene is the most influential compound with negative contributions. (B) Two reactions that only differ in reactant A have distinguished reaction yields. The benzene and chloride in 4-chloromethoxybenzene have negative contributions in low yield reaction prediction, while the iodide and pyridine ring in 2-iodopyridine show positive contributions in high yield reaction prediction.

most popular tools to decipher machine learning models via game theoretic approach. It measures the contributions to the final outcome (i.e., prediction) from each player (i.e., feature) separately among the coalition. For a particular prediction, every input token was assigned a SHAP value, which can be viewed as its contribution to that prediction. For a better visualization to reveal chemical insights, tokens are grouped as functional groups.

Figure 6 shows one reaction example from USP-TO_500_MT testset. It is a Pt catalyzed reduction. Input compounds for each task are colored based on their SHAP values using bwr colormap from matplotlib.⁸² Generally, SHAP values are calculated for each output token with regard to every input token. Here we map tokens back into atoms and clustered them into functional groups with EFGs package (<https://github.com/HelloJocelynLu/EFGs>).⁸³ In forward prediction and retrosynthesis, we summed the SHAP values over those changed atoms, as we would like to reveal contributions from inputs that lead to the key transformation. In reagents prediction, we examined on all output atoms since we want to have an overview for whole reaction environment.

As expected, the nitro group, as well as its reduced species—amine group, have strong positive contributions to this transformation in all three tasks. Interestingly, the aromatic ring also shows strong positive contributions in all tasks even though it is not directly involved in this reaction. Recall that nitro groups in alkyl and aryl nitro compounds are in different chemical environments, and behave differently, it may imply that our model learned to pay attention to key substructures even though they are not directly involved in a transformation. In forward prediction, the catalyst Pt also have positive contribution to product as expected. In retrosynthesis, we

noticed that chloride and another side chains also showed some SHAP intensities. It may due to the fact that this compound is also potentially synthesizable by chlorination or ether synthesis, but with much lower probability, as indicated by more predictions if we increased the beam size.

Figure 7 A) shows most influential reaction components in C–N coupling reactions.⁴³ The SHAP values indicate that the selections of aryl halides, additives and catalysts contribute most to all yield predictions. Among which 4-chloromethoxybenzene plays the most significant role in reaction yield prediction. We noticed that the benzene ring and chloride substitution have strong negative contributions. This observation can be demonstrated by the fact that the average reaction yield with 4-chloromethoxybenzene as one reactant is 30% lower than total average.

To show how SHAP value explanation can help us optimize reactions, we select two reactions with distinguished reaction yields as illustrated in Figure 7 B). These two reactions are only different on reactant A, aryl halides, and have the same reaction conditions. T5Chem predicts both reaction yields successfully. We noticed that the benzene and chloride in 4-chloromethoxybenzene have negative contributions to T5Chem predictions while the iodide and pyridine ring in 2-iodopyridine show positive contributions. This example implies that one may modify negative contributed substructures in reaction components to get a higher predicted yield.

CONCLUSION

In this work, we presented an explainable and unified transformer model, T5Chem, for multiple machine learning tasks related to organic chemistry synthesis. Our T5Chem shows state-of-the-art performances for four distinct types of

reaction prediction tasks using four different open-source data sets. Furthermore, we introduced a new data set USP-TO_500_MT for multitask machine learning of chemical reactions, including forward reaction prediction, retrosynthesis, reagents prediction, reaction type classification (500 classes), and reaction yield prediction. Our results showed that T5Chem models trained with multiple tasks are more robust and can benefit from mutual learning on related tasks. Finally, we demonstrated the applicability of SHAP to explain T5Chem predictions at the functional group level, which provides a way to demystify sequence-based deep learning models in chemistry. This sets the stage to develop T5Chem into a widely applicable and versatile machine learning framework for a variety of prediction tasks in molecular science.

DATA AND SOFTWARE AVAILABILITY

T5Chem source codes and data sets are accessible through <https://yzhang.hpc.nyu.edu/T5Chem>.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.1c01467>.

Figures S1 and S2: Data distribution on USP-TO_500_MT. Table S1: Results for different tokenizations on USP-TO_500_MT. Figure S3: Learning curve for pretrained model vs model trained from scratch. Table S2: Results for multitask training using combined loss functions. Table S3: Results for data augmentation for sequence-to-sequence tasks. (PDF)

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Notes

The authors declare no competing financial interest.

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